

Class Notes: Phylum Mollusca

Animal Kingdom • PU-I Biology • Class Notes

Typical Molluscan Body Plan

second largest phylum in Animalia among invertebrates; terrestrial or aquatic; bilaterally symmetrical, triploblastic, coelomate, organ system level. Body covered by a calcareous shell with distinct head, muscular foot and visceral hump; a soft mantle covers the visceral hump. Mantle cavity (space between hump and mantle) contains feather-like gills with respiratory and excretory functions.

Physiology and Examples

Excretory organs: one or two pairs of kidneys called metanephridia; sensory tentacles on the head; feeding organ is the file-like radula. Molluscs are dioecious and oviparous, with direct or indirect development.

Examples: Pila (apple snail), Pinctada (pearl oyster), Octopus (devil fish), Sepia (cuttlefish), Loligo (squid), Aplysia (sea-hare), Dentalium (tusk shell), Chaetopleura (chiton).